

- 1) Apply Columnar Transposition cipher Method and encrypt the message "The tomato is a plant in the nightshade family" using keyword Network ignore space add assume padded character z?

Given encrypted message: The tomato is a plant in the nightshade family"

Keyword: Network.

After split message: 'Th', 'et', 'om', 'at', 'oi', 'sa', 'pl', 'an', 'ti', 'nt', 'he', 'ni', 'gh', 'ts', 'ha', 'de', 'fa', 'mi', 'ly'

arrange in matrix:

1) Th: [WG]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

2) ET: [TW]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

3) OM: [EB]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

4) AT: [GA]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

5) OI: [NW]  
CO

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

6) SA: [PE]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

7) PI: [MQ]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | X | Y | Z |   |

8) AN: [RT]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

9) TI: [OG]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

10) NT: [EW]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

11) HG: [FW]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

12) NI: [OD]

|   |   |   |   |   |
|---|---|---|---|---|
| N | E | T | W | O |
| R | K | A | B | C |
| D | F | G | H | I |
| L | M | P | Q | S |
| U | V | X | Y | Z |

Final ciphertext: [WGTAEAGACOPCQMRTOGIEWFWDDHIDPGBFNGKSFQU]

2) Analyse playfair cipher algorithm for given key: MONARCHY & Message: Instrument.

Sol:

Given key: MONARCHY

Message: instrument

After split: 'in', 'st', 'ru', 'me', 'nt', 'sz'

• Here  $z$  is padded to make even number of characters.

• ciphertext: gatlmzclorgtz

1) IN:

|   |   |   |   |   |
|---|---|---|---|---|
| M | O | N | A | R |
| C | H | Y | B | D |
| E | F | G | I | K |
| L | P | Q | S | T |
| U | V | W | X | Z |

2) ST:

|   |   |   |   |   |
|---|---|---|---|---|
| M | O | N | A | R |
| C | H | Y | B | D |
| E | F | G | I | K |
| L | P | Q | S | T |
| U | V | W | X | Z |

3) RU:

|   |   |   |   |   |
|---|---|---|---|---|
| M | O | N | A | R |
| C | H | Y | B | D |
| E | F | G | I | K |
| L | P | Q | S | T |
| U | V | W | X | Z |

4) ME:

|   |   |   |   |   |
|---|---|---|---|---|
| M | O | N | A | R |
| C | H | Y | B | D |
| E | F | G | I | K |
| L | P | Q | S | T |
| U | V | W | X | Z |

5) NT:

|   |   |   |   |   |
|---|---|---|---|---|
| M | O | N | A | R |
| C | H | Y | B | D |
| E | F | G | I | K |
| L | P | Q | S | T |
| U | V | W | X | Z |

6) SZ:

|   |   |   |   |   |
|---|---|---|---|---|
| M | O | N | A | R |
| C | H | Y | B | D |
| E | F | G | I | K |
| L | P | Q | S | T |
| U | V | W | X | Z |

3) Apply Caesar Cipher and  $k=5$  decrypt the given ciphertext "YMITYMIWXNIITTKXNQISHI"

Sol:

Given Cipher text: "YMITYMIWXNIITTKXNQISHI"

given key  $k=5$

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 |
| A  | B  | C  | D  | E  | F  | G  | H  | I  | J  | K  | L  | M  | N  |
| O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |    |    |
| 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |    |    |

We know that Decrypt formula

$$p = (c - k) \bmod 26$$

# step-by-step

| c | Value | (c-5) % 26 | p  | Letter |
|---|-------|------------|----|--------|
| Y | 24    | 19         | 19 | T      |
| M | 12    | 7          | 7  | H      |
| J | 9     | 4          | 4  | E      |
| T | 19    | 14         | 14 | O      |
| Y | 24    | 19         | 19 | T      |
| M | 12    | 7          | 7  | H      |
| J | 9     | 4          | 4  | E      |
| W | 22    | 17         | 17 | R      |
| X | 23    | 18         | 18 | S      |
| N | 13    | 8          | 8  | I      |
| I | 8     | 3          | 3  | D      |
| J | 9     | 4          | 4  | E      |
| T | 19    | 14         | 14 | O      |
| K | 10    | 5          | 5  | F      |
| X | 23    | 18         | 18 | S      |
| N | 13    | 8          | 8  | I      |
| Q | 16    | 11         | 11 | L      |
| J | 9     | 4          | 4  | E      |
| S | 18    | 13         | 13 | N      |
| H | 7     | 2          | 2  | C      |
| J | 9     | 4          | 4  | E      |

plain Text: THE OTHER SIDE OF SILENCE

4) Does the SSL protocol need the Service of a Certificate authority? Explain your answer?

Yes, SSL protocol requires the Services of a Certificate Authority. A Certificate authority (CA) is a trusted Organization that issues digital Certificates to websites, Verify their identity. When the user Connects to a Secure Website using SSL, the Website presents its Certificate, which is signed by a CA, the user's browser checks this Certificate to make sure the website is genuine and not fake. This helps prevent attacks like impersonation and ensure Secure Communication. Without a CA, users would not be able to trust whether a website is legitimate, making SSL Connection less Secure.

5) Christopher wants to send the data ( $M=9$ ) to John Lisa in a Confidential manner. Analyze the RSA encryption process used by Christopher, assuming  $p=3$ ,  $q=11$  and  $e=7$ . Also assist John Lisa in decrypting the ciphertext and justify the process compared to symmetric key encryption!

### RSA Encryption & Decryption

Christopher wants to send the message ( $M=9$ ) Securely using RSA

He selects two prime numbers:

$$p = 3, q = 11$$

First, Compute

$$n = p \times q = 3 \times 11 = 33$$

$$\phi(n) = (p-1)(q-1) = 2 \times 10 = 20$$

He chooses public Content

$$e = 7$$

Now, find private key  $d$  such that

$$ed \equiv 1 \pmod{20}$$

$$d = 3 \left( \because 4 \times 3 = 12 \equiv 1 \pmod{20} \right)$$

So

$$\cdot \text{Public key} = (4, 33)$$

$$\cdot \text{Private key} = (3, 33)$$

Encryption (by Christopher)

$$C = M^e \pmod{n} = 9^4 \pmod{33}$$

$$9^2 = 81 \equiv 15 \pmod{33}$$

$$9^4 \equiv 27 \pmod{33}$$

$$9^4 = 9^2 \cdot 9^2 \equiv 15 \cdot 15 \equiv 225 \equiv 27 \pmod{33}$$

$$\text{Ciphertext} = 27$$

Decryption

$$M = C^d \pmod{n} = 27^3 \pmod{33}$$

$$27^2 = 729 \equiv 27 \pmod{33}$$

$$27^3 = 27 \cdot 27 = 729 \equiv 27 \pmod{33}$$

$$\text{Decrypted message} = 27$$